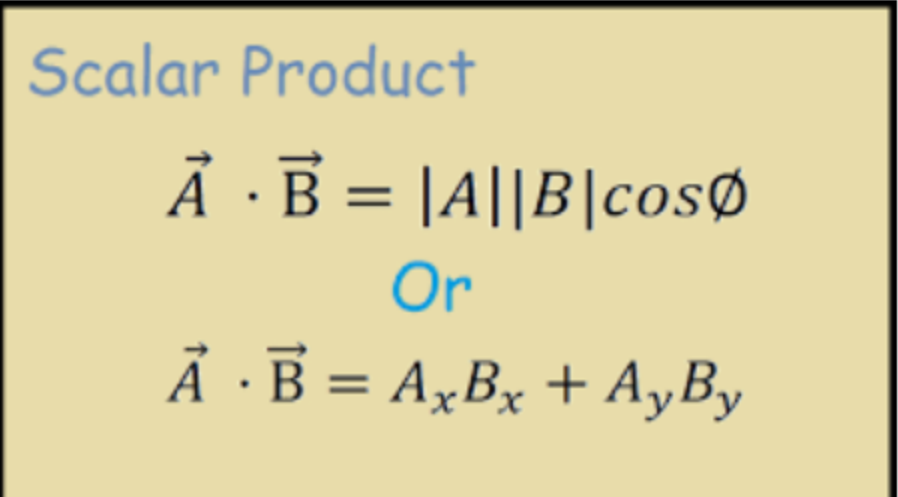


7_ popquiz Gauss's law bonus ahead ↕

⚠ This is a preview of the published version of the quiz

Started: Mar 15 at 3:12pm

Quiz Instructions



Scalar Product

$$\vec{A} \cdot \vec{B} = |A||B|\cos\theta$$

Or

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$$

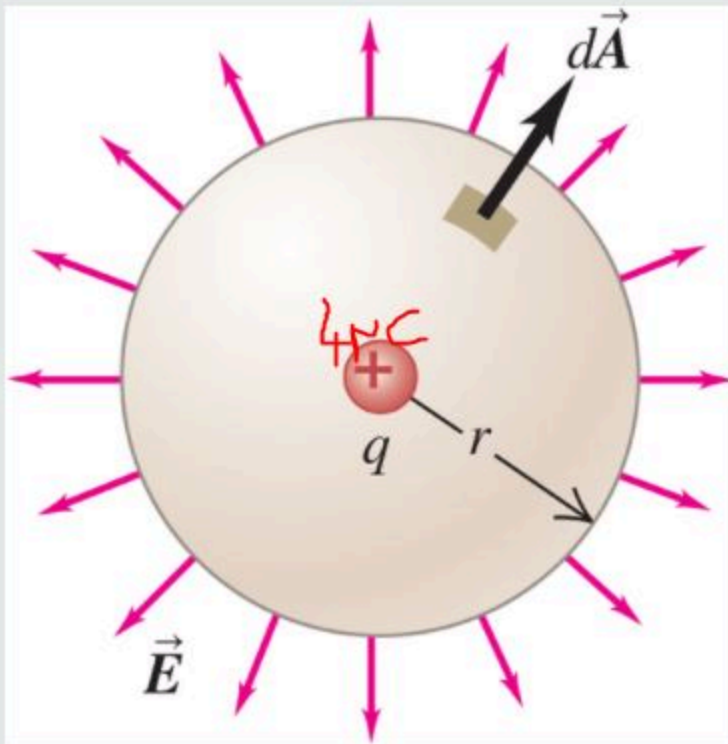
And micro is 10^{-6} and nano 10^{-9}



Question 1 20 pts

A point charge $q = +4.5 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge (Fig. 22.9). Find the resulting electric flux through the sphere.

FIGURE 22.9



hint: Just apply Gauss's law: flux = Q_{in}/ϵ_0

μC means micro coulomb

ϵ_0 check your equation sheet. vacuum permittivity. (epsilon 0)

☐ $4.5 \cdot 10^5 \text{ Nm}^2/\text{C}$

☐ $2 \cdot 10^3 \text{ Nm}^2/\text{C}$

☐ $3.4 \cdot 10^5 \text{ Nm}^2/\text{C}$

☐ $3.4 \cdot 10^4 \text{ Nm}^2/\text{C}$



Question 2 17 pts

The flux of a vector field $\vec{E}$ through a surface A depends on:

(mathematically. Think of a net catching moving shrimps)

☐ only on the orientation of A

☐

only on the strength of E

☐

only on the area A

☐

the strength of E, area A and orientation of A



Question 3 17 pts

A large aluminum foil is charged with a van graaf (like in the video I showed you).

Its electric field at the surface is 10,000N/C

The dimension of the foil is 2m x 4m. What is the charge surface density ?

hint: again. again. in your first equation sheet you have the equation of the electric field close to a large surface. *Esheet* the notation is.

(use the constant ϵ_0 vacuum permittivity)

☐

8.9 10^{-8} C/m²

☐

1.8 10^{-7} C/m²

☐

2.6 10^{15} C/m²

☐

8.85 10^{-12} C/m²



Question 4 20 pts

Building on the previous question. What is charge Q on the sheet?

1uC means micro coulomb

1nC means nano coulombs

☐

0.7 uC

☐

1.4 uC

☐

0.4 uC

☐

1.4 nC



Question 5 17 pts

Consider a dipole with an electric moment p inside an electric field.

$p = (3\mathbf{i} + \mathbf{j}) \text{ Cm}$ (hint: its a vector notation so $x=3$ and $y=1$ or $p(3,1)$)

$E = (4\mathbf{j}) \text{ N/m}$ (hint: its a vector notation so $x=0$ and $y=4$ or $E(0,4)$)

What is its potential energy U

hint: check the definition of scalar product (dot product) in the header. and slide 103 for the equation for U

or equation sheet

☐

-13J

☐

-12J

☐

-16 J/

☐

-4 J

☐

Question 6 20 pts

building on previous question.

Draw the vector p (you can attach its tail at the origin). The tail has a negative charge and the head has a positive charge. since the vector p goes from - to +

draw a field of E . that is vertical vectors pointing up $E(0,4)$. p is inside that vector field.

What is true ?

remember there is a torque + want to go with the E flow.

☐

The dipole wants to rotate CCW

☐

the dipole wants to rotate CW

☐

whats a dipole

☐

the dipole does not move

Not saved

Submit Quiz